CS 490: Goodreads Genre Network

Analysis of Book Similarity Networks

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Introduction

Goal: Build a book recommendation network by clustering Goodreads books based on genre similarity rather than Goodreads' native categorization

Methods Used:

- Jaccard similarity for genre matching (weight: 0.6)
- Number of ratings similarity (log-scaled, weight: 0.2)
- Rating similarity (1 - —r1-r2— / 1.0, weight: 0.2)
- Combined: $0.6 \times$ genre $+ 0.2 \times$ num_ratings $+ 0.2 \times$ rating
- Threshold $= 0.7$ for edge creation

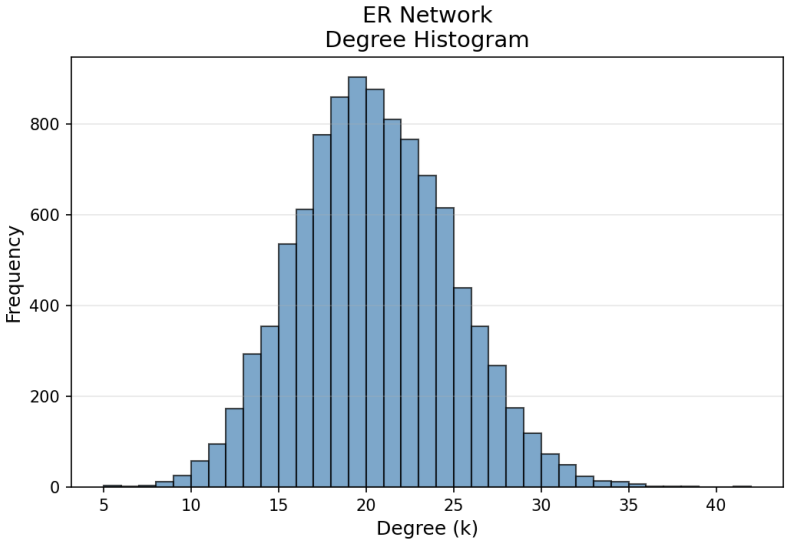
Data: 10,000 books from Kaggle datasets

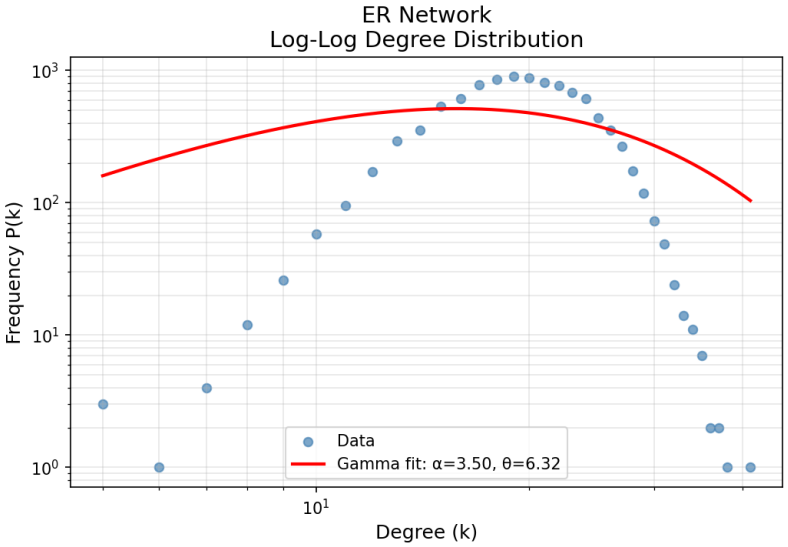
Basic Network Statistics

Metric	Goodreads Network	ER Network
Nodes	10,000	10,000
Edges	175,890	100,000
Density	0.0035	0.0020
Avg Degree	35.18	20.00
Max Degree	436	41

Random Graph (Erdôs-Rényi)

Generated ER random graph with same n,m as our network



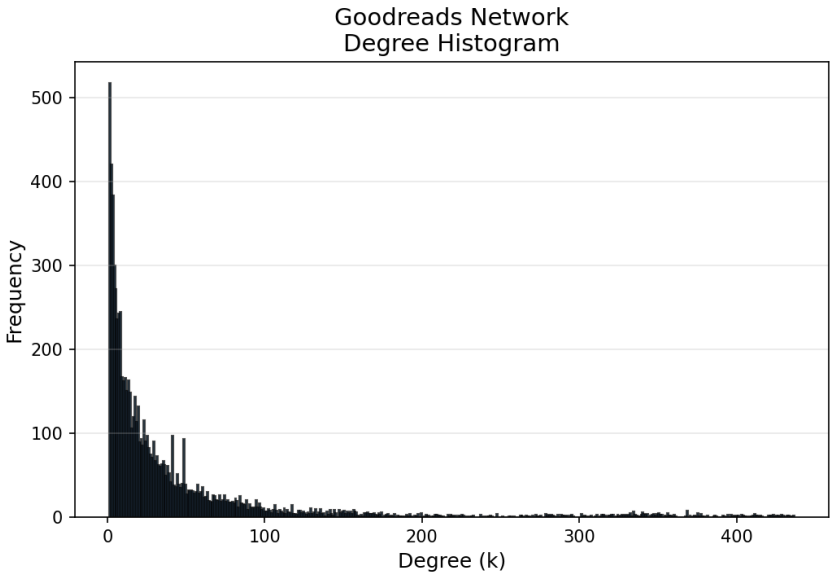


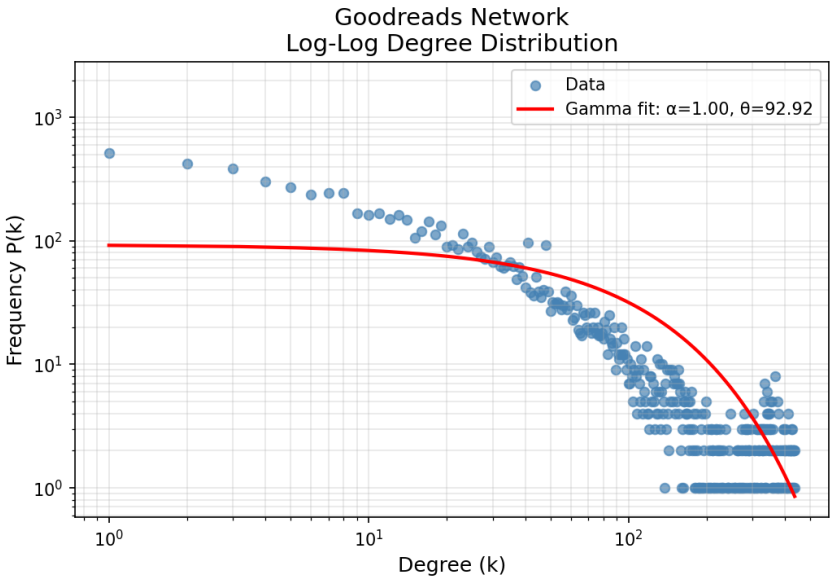
Metric	Goodreads Network	ER Network
Clustering Coeff	0.4137	0.0020
Modularity (Greedy)	0.7103	0.1740
Modularity (Louvain)	0.7686	0.1795
Assortativity	0.7801	0.0033

Gephi Graph Visualization

Color coding: By community / genre cluster
Size: Node size by degree

Degree Distribution





Key observations:

- Heavy-tailed distribution (scale-free properties)
- Few highly connected "hub" books connect many genres
- Average degree: 35.18 vs ER average: 20.00

Graph Analysis Methods

Statistics Computed:

- Nodes, Edges, Density
- Degree distribution (mean, variance, max)
- Clustering coefficient
- Degree assortativity

Community Detection:

- Greedy Modularity
- Louvain method
- Label Propagation

All computed using NetworkX library

Comparison with Original Goodreads Network

Our approach vs Goodreads' native genre categorization:

Feature	Our Network	Goodreads
Clustering basis	Genre + Num_Ratings + Rating	Single genre
Similarity metric	Jaccard + log(popularity) + Rating	Binary
Weights	0.6 + 0.2 + 0.2	Implicit
Edge threshold	0.7	Implicit
Community detection	Yes (Q=0.77)	No

Note: Need to research Goodreads' actual method

Clustering Coefficient Explained

Measures degree of transitivity in the network

$C = \frac{3 \times \text{triangles}}{\text{connected_triples}}$

Interpretation:

- $C \approx 0$: Random network
- $C > 0.01$: Some clustering (our expected result)
- $C > 0.1$: Highly clustered

Community Detection Methods Comparison

Method	Communities	Modularity	Runtime
Greedy Modularity	2,101	0.7103	56s
Louvain	2,080	0.7686	4s
Label Propagation	2,327	0.7383	2s

Quality metric: Modularity $Q \in [-0.5, 1]$

- $Q > 0.3$: Significant community structure
- Higher = better partitioning

Interpretation of Results

Key Findings:

1. Books cluster naturally by genre $\rightarrow$ strong modularity
2. Degree distribution follows power-law $\rightarrow$ scale-free
3. Comparison with ER shows significant structure

Implications:

- Genre-based similarity captures real book relationships
- Network analysis reveals book communities beyond single-genre labels
- Potential for better recommendation systems

Questions?

Code: github.com/zaigiaz/goodreads-network

Run: python data_analysis.py –file 1 –clustering –communities –assortativity